How heat and fuel become motion

Machines transform energy
Not magic. Real engineering.

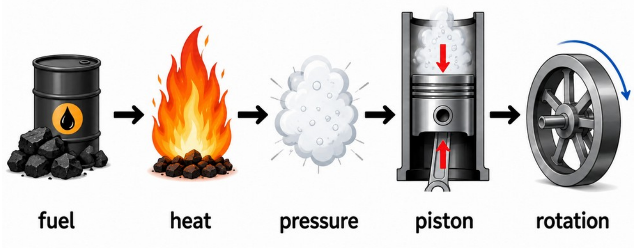
Useful work
Force + movement

Energy losses
Heat, sound, friction

Today we connect mechanics with engines.

We will study: work, power, efficiency, heat, pressure, steam engine, combustion engine.

We will also solve simple engineering problems.



From Simple Machines to Engines

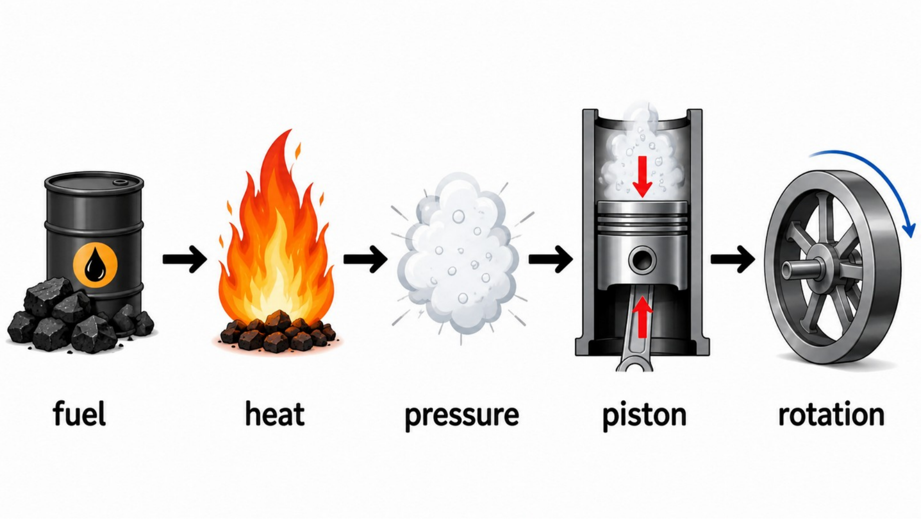
Review and new question

Simple machines help us use force.

But simple machines do not create energy.

Real machines need an energy source.

Today's main question: How does energy become motion?



lever

wheel

pulley

ramp

wedge

screw

→ then engines

Work

Force with movement

Mechanical work needs two things:

- force
- movement

Push a wall: no movement → no useful work.

Push a bed: movement → work is done.

$$W = F \times s$$

W: work (J), F: force (N), s: distance (m)

1 joule = 1 newton × 1 meter

Work is energy transfer by force.

Work Example

Very simple numbers

$F = 60$ N
$s = 3$ m

$W = F \times s$

$W = 60 \times 3 = 180$ J

Example: a nurse pushes a hospital bed.

Force = 60 N.

Distance = 3 m.

Work = 180 J.

Student task
$F = 40$ N, $s = 2$ m
Work = ?
Answer: 80 J

Energy

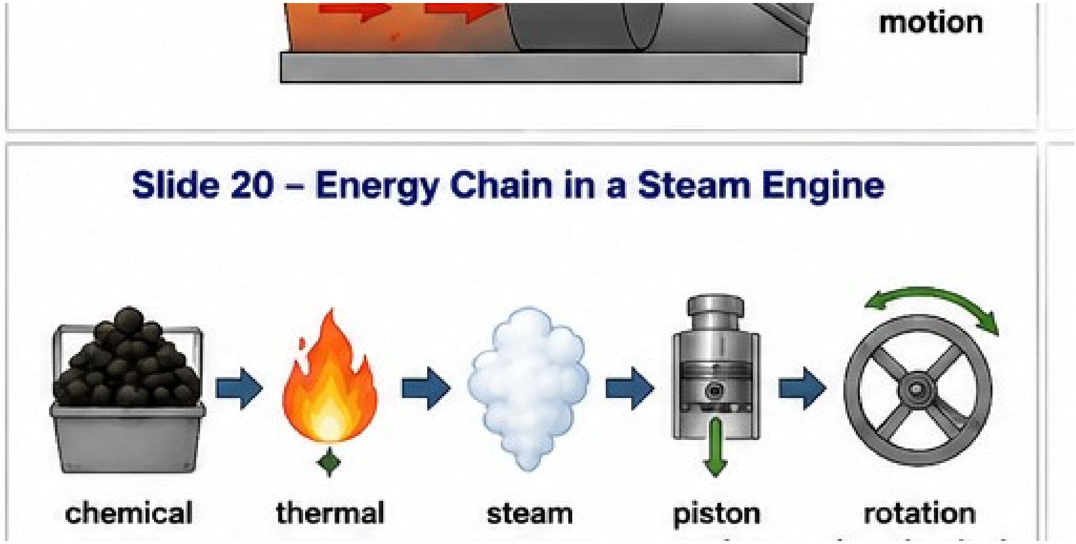
Ability to do work

Energy means ability to do work.

Unit: joule (J).

Machines do not create energy.

Machines transform energy from one form to another.



chemical energy → fuel

thermal energy → heat / steam

mechanical energy → motion

electrical energy → next lecture

Power

How fast work is done

Two machines can do the same work.

The faster machine has more power.

Same work + less time = more power.

$$P = W / t$$

P: power (W), W: work (J), t: time (s)

Unit: watt

1 watt = 1 joule per second

Power Example

Same work, different time

Work = 180 J
Time = 6 s

$P = W / t$

$P = 180 / 6 = 30$ W

Power tells us the speed of doing work.
Bigger power means quicker work.

Student task
W = 200 J, t = 10 s
Power = ?
Answer: 20 W

Efficiency

Mechanisms and machines lose energy

No real machine is perfect.

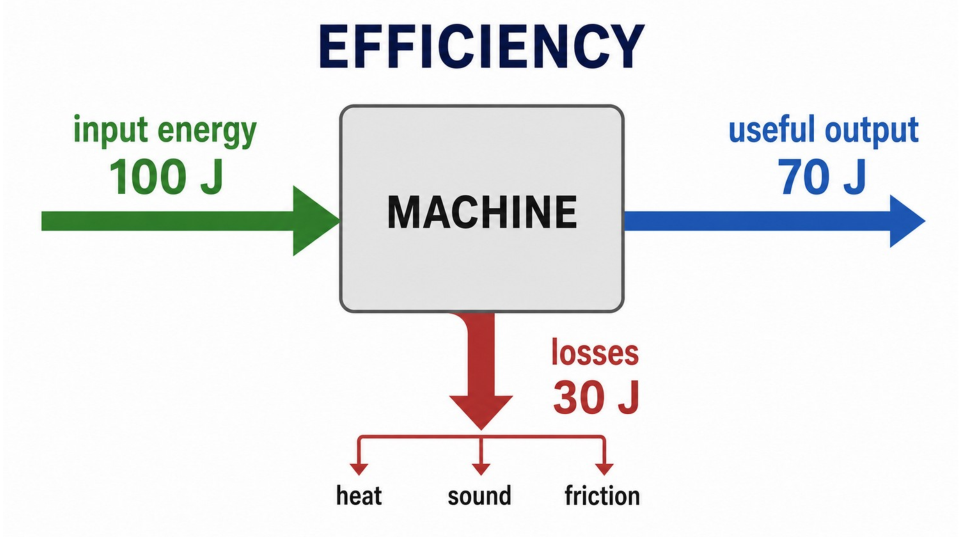
Some input energy becomes useful output.

Some energy is lost as heat, sound, vibration, friction.

Efficiency shows useful part of the input.

$$\eta = \text{useful output} / \text{input} \times 100\%$$

Efficiency is usually written in percent.



High efficiency = small losses

Low efficiency = big losses

Efficiency of One Mechanism

Warm-up calculation

Input = 100 J

Useful output = 80 J

$\eta = 80/100 \times 100\% = 80\%$

Example: a belt or gear mechanism.

20 J is lost.

Useful part = 80 J.

Efficiency = 80%.

Student task

Input = 200 J

Output = 150 J

Efficiency = ?

Answer: 75%

Percent to Decimal

Very important for efficiency calculations

To multiply efficiencies, first change percent to decimal.

Rule: divide by 100.

90% = 0.90

80% = 0.80

75% = 0.75

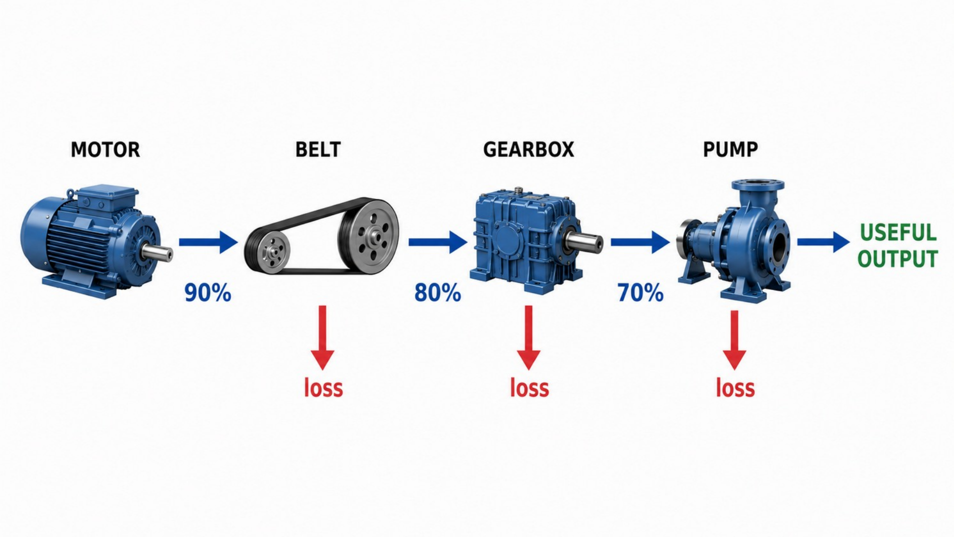
50.4% = 0.504

Percent → decimal
90% ÷ 100 = 0.90

Decimal → percent
0.72 × 100 = 72%

Multi-Stage Efficiency

Many small losses become one big loss



Each stage loses some energy.

Total efficiency gets smaller and smaller.

$\eta_{total} = \eta_1 \times \eta_2 \times \eta_3$

Use decimal numbers: 0.90, 0.80, 0.70

Example

$0.90 \times 0.80 \times 0.70 = 0.504$

$\eta_{total} = 50.4\%$

Useful Output Power

Input power and total efficiency

Input power = 500 W
ηtotal = 60% = 0.60

Useful output power
= 500 × 0.60 = 300 W

Useful output power = input power × efficiency.

Always change percent to decimal first.

Student task
Input = 100 W
η = 72%
Output = ?
Answer: 72 W

Heat and Temperature

They are not the same

Temperature
How hot something is

Heat
Energy moving from hot object to cold object

A hot engine gives heat to air.

Hot steam gives heat to metal parts.

Heat flow is important in all heat engines.

Gas Particles and Temperature

Why hot gas makes pressure

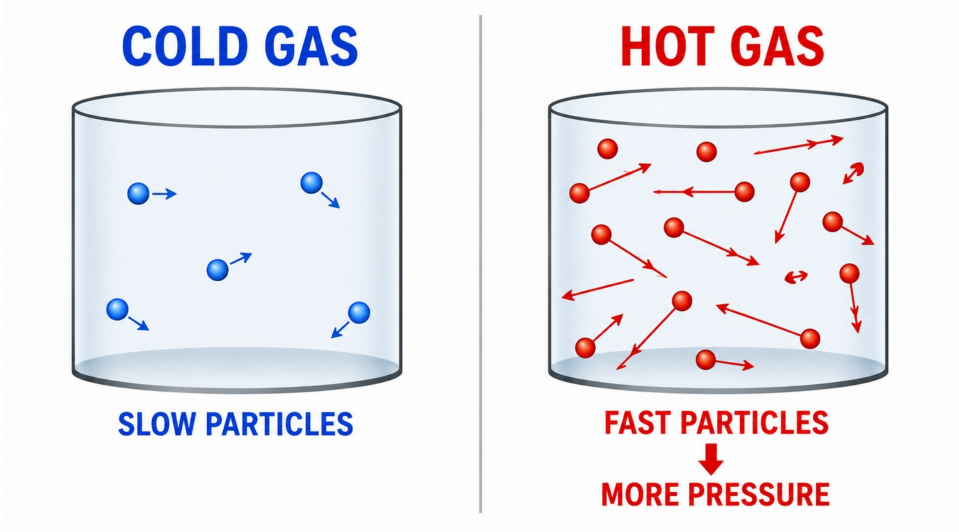
Gas is made of tiny moving particles.

Cold gas: particles move slowly.

Hot gas: particles move faster.

Fast particles hit the walls harder.

This gives more pressure.



higher temperature

→ faster particles

→ bigger pressure

Pressure

Force on area

Pressure is very important in engines.

Pressure can push a piston.

Two useful formulas:

$$P = F / A$$

$$F = P \times A$$

P: pressure (Pa), F: force (N), A: area (m^2)

bigger pressure → bigger force

bigger piston area → bigger force

Pressure Example

Pressure makes force

Pressure = 100,000 Pa
Area = 0.01 m^2

$F = P \times A$

$F = 100{,}000 \times 0.01 = 1{,}000$ N

Even a small piston can make big force if pressure is high.

This is the basic idea of steam and combustion engines.

Student task

P = 200,000 Pa

A = 0.02 m^2

Force = ?

Answer: 4,000 N

Gas Expansion

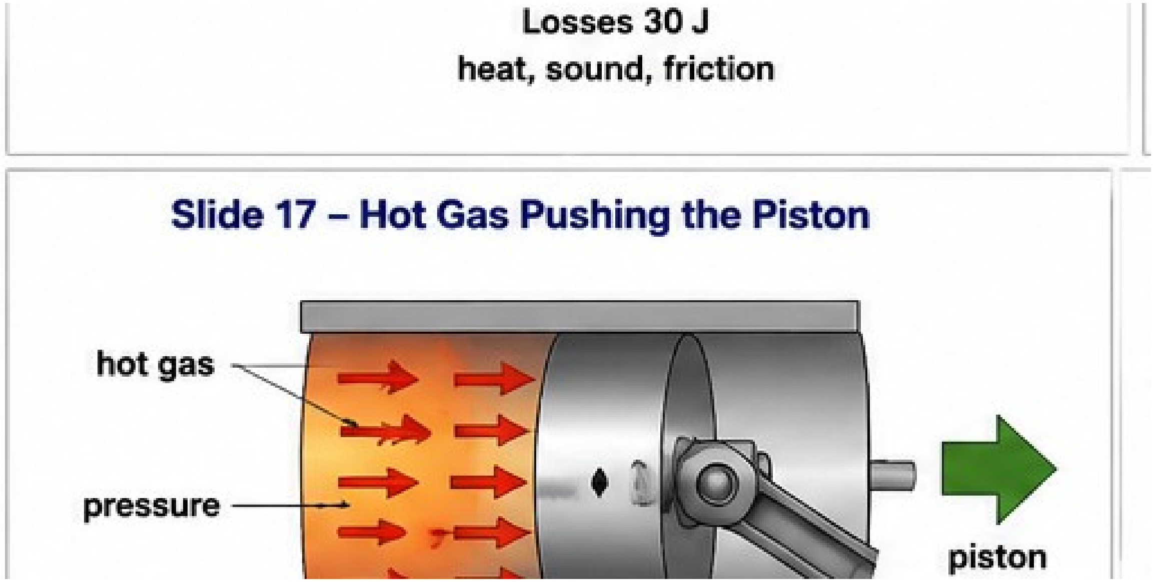
Hot gas wants to take more space

When gas becomes hot, it expands.

Expand means it tries to become bigger.

Inside a cylinder, gas cannot go everywhere.

So it pushes the piston.



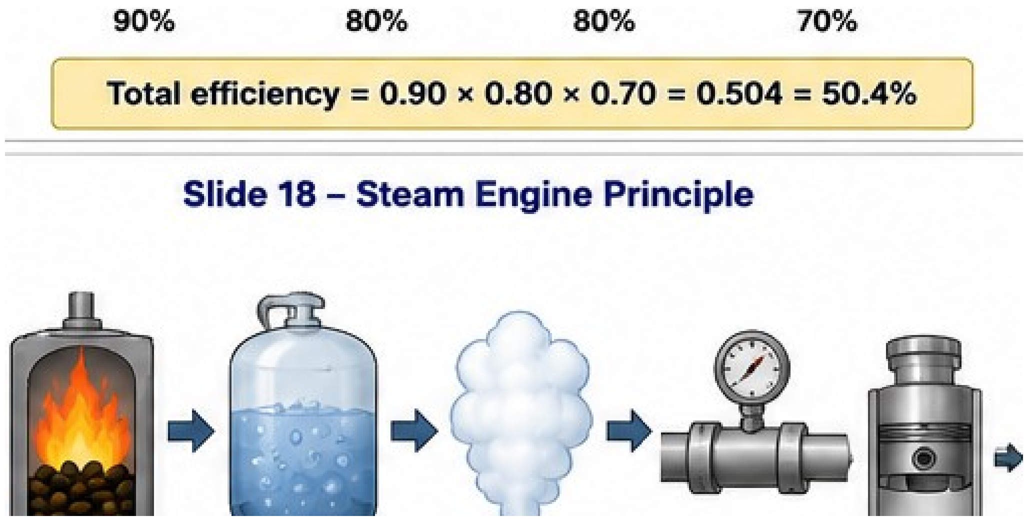
hot gas expands

expansion pushes piston

piston motion can do work

Steam Engine Principle

From fire to motion



Fire heats water.

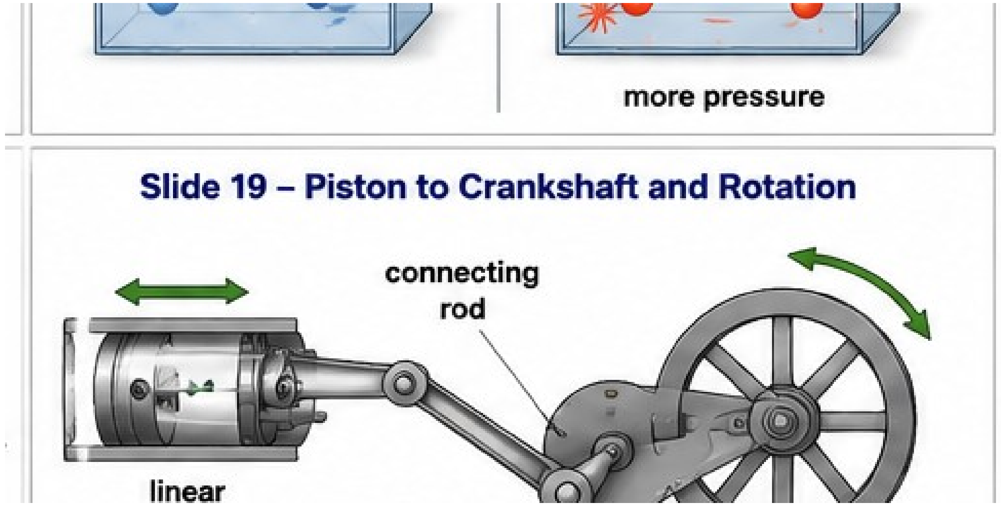
Water becomes steam.

Steam pressure pushes the piston.

The piston moves and can do useful work.

From Piston to Rotation

Straight motion becomes rotation



Piston motion is straight line motion.

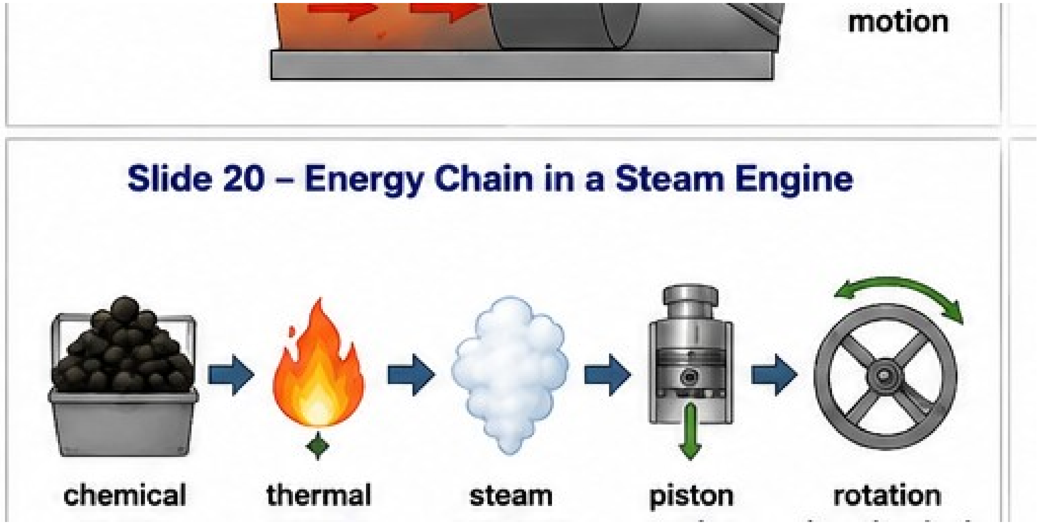
Many machines need rotation.

A connecting rod and crankshaft change straight motion into rotation.

Steam engines and combustion engines both use this idea.

Steam Engine Energy Chain

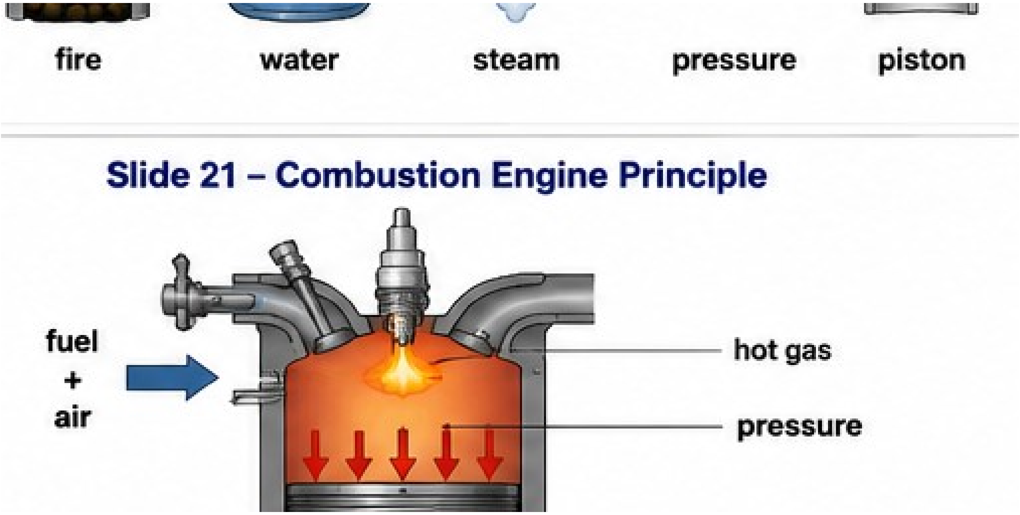
Follow the energy step by step



Fuel has chemical energy.
Burning gives heat.
Heat makes steam.
Steam pressure pushes the piston.
Piston motion becomes rotation.

Internal Combustion Engine

Fuel burns inside the cylinder



Fuel and air enter the cylinder.

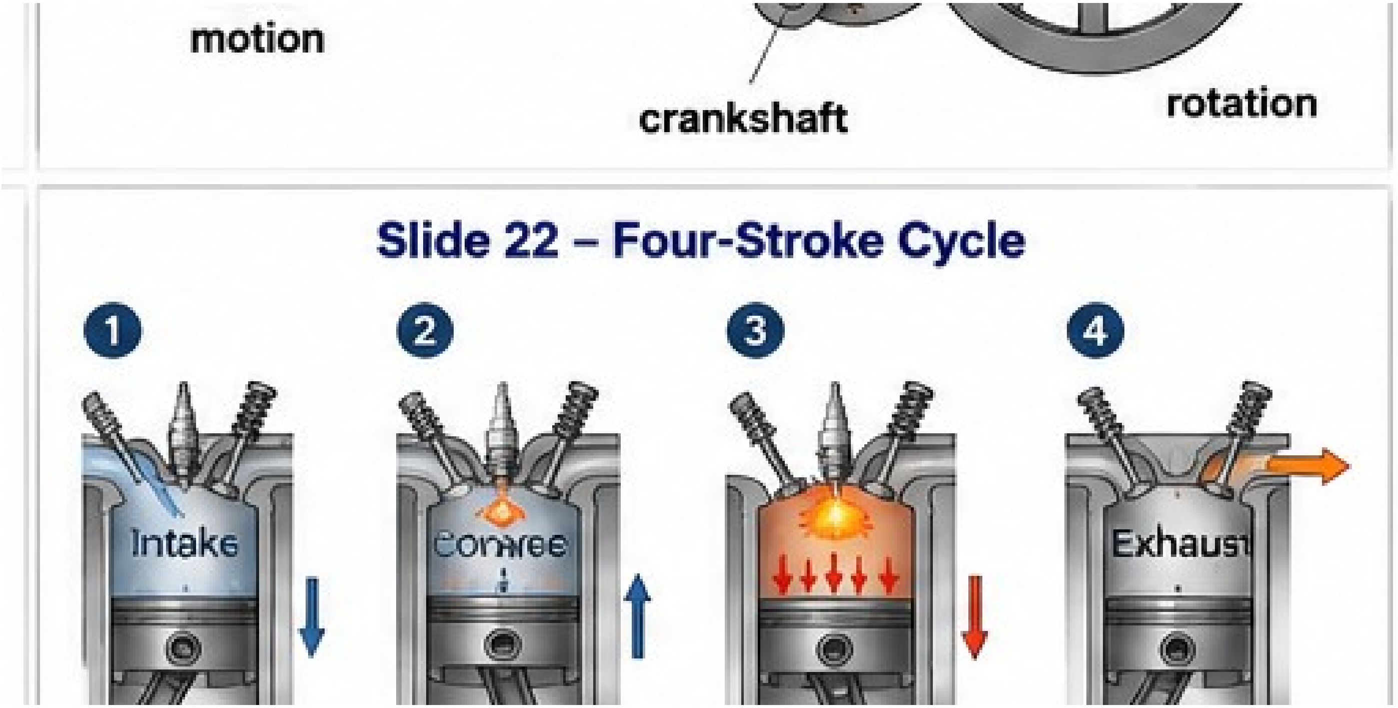
Burning makes hot gas.

Hot gas pressure pushes the piston.

The crankshaft changes piston motion into rotation.

Four Strokes

Basic cycle of a gasoline engine



This cycle repeats again and again to make continuous rotation.

Steam Engine vs Combustion Engine

Important comparison

Steam engine
Burning is outside the cylinder.
Water becomes steam.
Steam pushes the piston.

Combustion engine
Burning is inside the cylinder.
Fuel burns in the cylinder.
Hot gas pushes the piston.

Both engines use pressure, piston motion, and crankshaft rotation.

Gasoline and Diesel

Short comparison

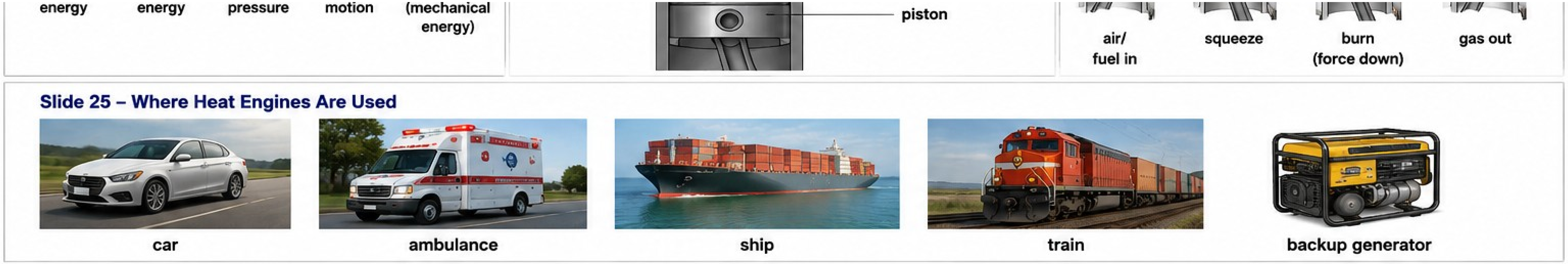
Gasoline engine	Diesel engine
A spark starts burning.	High compression starts burning.

Both are combustion engines.

Both turn chemical energy in fuel into mechanical motion.

Where We Use Heat Engines

Useful engineering examples



cars
ambulances
ships
trains
construction machines
backup generators
Hospital emergency power often depends on a fuel engine + generator.

Troubleshooting Idea

If an engine does not work, ask simple questions first

1. Is there fuel?
2. Is there air?
3. Is there ignition?
4. Is there compression?
5. Can the moving parts move?
6. Is something blocked or broken?

Calculation Practice

Short tasks for students

1. Work: $F = 50$ N, $s = 4$ m. $W = ?$
2. Power: $W = 200$ J, $t = 10$ s. $P = ?$
3. Efficiency: input = 500 J, useful output = 150 J. $\eta = ?$
4. Pressure force: $P = 200{,}000$ Pa, $A = 0.02\ m^2$. $F = ?$
5. Multi-stage efficiency: 90% and 80%. $\eta total = ?$
6. Useful output power: input = 100 W, $\eta = 72\%$. output = ?

Answers: 200 J, 20 W, 30%, 4,000 N, 72%, 72 W

Summary

Main ideas to remember

Work = force × distance.

Power = work / time.

Efficiency = useful output / input × 100%.

For multi-stage machines, multiply efficiencies in decimal form.

Heat raises temperature. Hot gas makes pressure.

Pressure pushes the piston.

Steam engine uses steam pressure.

Combustion engine uses hot gas pressure.

Crankshaft changes straight motion into rotation.

Teacher Checklist: Images to Generate

Use this slide for preparation only

- S1 banner chain: fuel → heat → pressure → piston → rotation
- S8 input energy → useful output + losses
- S11 motor → belt → gearbox → pump
- S14 cold gas vs hot gas particles
- S17 hot gas pushing piston in cylinder
- S18 steam engine process diagram
- S19 piston → connecting rod → crankshaft → wheel
- S20 steam engine energy chain
- S21 combustion engine process diagram
- S22 four-stroke cycle diagram
- S25 collage: car, ambulance, ship, train, backup generator